

Steve Cabrel Teguia Kouam
Department of Physics, Faculty of Science
University of Douala, Cameroon
steve.teguia@facsciences-uy1.cm

September 25, 2026

Editorial Office
Applied Energy
Elsevier

Dear Editor,

We are pleased to submit our manuscript entitled “**Quantum Agrivoltaic Digital Twin: Coordinated Vibronic Light-Harvesting, Environmental Calibration, and Agro-Rural Cybersecurity**” for consideration as a Research Article in *Applied Energy*. It is not under consideration elsewhere; the journal’s systems-level energy–agriculture–water scope is its natural home.

Significance of the work

Agrivoltaic deployment is constrained by a system-level trade-off: static semi-transparent photovoltaic modules resolve the food–energy land conflict only by uniformly sacrificing either electrical output or crop productivity. Using numerically exact quantum dynamics (PT-HOPS/SBD), we show that an organic photovoltaic layer integrated with a nanoparticle-on-mirror (NPoM) plasmonic cavity can *actively* shape the transmitted spectrum to protect quantum coherence in the photosynthetic machinery, while the same cavity enables label-free SERS diagnostics of crop health. The digital twin links these quantum effects to canopy yields, Net Ecological Benefit, and cooperative economics.

Key advances

- **Coherence protection:** dual-band filtering at 750 nm and 820 nm extends exciton coherence by 50 % and raises the forward transfer yield by 25 % at room temperature.
- **Experimental testability:** all modelled observables are anchored to independent experimental benchmarks, with two validation protocols (dual-band 2DES with SLM pulse shaping; time-resolved NPoM SERS).
- **SERS–transport trade-off resolved:** the NPoM diverts 79–84 % of the excitation to the plasmon mode, but confined to a 1 % sentinel area, the 91.8 % local transport penalty costs 2.9 canopy-yield points (97.1 % preserved) while adding single-complex SERS diagnostics to the same footprint.
- **Farm-scale proof of concept:** a 500 m² cooperative greenhouse in Cameroon combines 90 MWh yr^{−1} of OPV electricity, 28 % evapotranspiration savings, a Net Ecological Benefit of 19.6 kg_{CO₂e} m^{−2} yr^{−1}, and a 4.23 yr payback under 30 % subsidy.

Why *Applied Energy*?

The journal’s readership spans photovoltaics, agricultural-energy systems, the water–energy–food nexus, lifecycle assessment, and energy economics — precisely the scales this work integrates.

Novelty, compliance, and declarations. The integration of spectral coherence protection, NPoM SERS diagnostics, carbon-financed cooperative economics, and QKD-secured IoT in one platform is, to our knowledge, unreported; specifically, we report the first (i) agrivoltaic stack whose spectral architecture is co-designed by exact open-quantum-system dynamics, jointly optimizing excitation transport and power conversion, (ii) monolithic photothermal–PV–SERS device concept with field-relevant calibration in a single footprint, and (iii) multifunctional techno-economic and lifecycle framework internalizing heat, crop, water, sensing, and data-security trade-offs (\$/W + \$/kg + \$/m³). Deployment barriers are made explicit: economic (tractable — local materials halve CAPEX) versus physical (nanofabrication, fouling, encapsulation lifetime). The manuscript has been checked by a native English speaker with expertise in the field; all authors approved it, declare no competing interests, and are available to review at least three other *Applied Energy* articles this year. An earlier version was considered at *Nature Energy*; we submit directly here as the more suitable venue. Subscription publication (no APC due); the corresponding author is based in Cameroon.

Suggested reviewers: A. Olaya-Castro (UCL), J. Yuen-Zhou (UCSD), J. J. Baumberg (Cambridge), G. A. Barron-Gafford (Arizona) — covering quantum biology, polaritonic chemistry, NPoM/SERS, and agrivoltaics.

Sincerely,

Steve Cabrel Teguia Kouam (on behalf of all authors)

Enclosures: Manuscript, Supplementary Material (SI.pdf), Highlights, Graphical Abstract.